Course Registration System Redesign
Vision and Scope

Issy Chen, Ian Hutchings, Winson Teh, Ashley Vo
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Solution Vision

Global State University (GSU) struggles to provide an efficient and accessible course registration system for students and staff. The current system is fragmented, lacks real-time updates, and is unable to handle peak registration traffic, leaving students, advisors, and administrators frustrated and at risk of errors and compliance issues.

Solution Vision

The proposed solution is a modern, cloud-based registration system designed to provide real-time course information, intelligent scheduling tools, and integrate with existing university systems, including the Student Information System, Learning Management System, Financial Aid System, and Degree Audit System.

The system will offer mobile-responsive and accessible interfaces with multi-language support and built-in dashboards. Primary features will include automated prerequisite checking, waitlist management with real-time notifications, and integrated degree progress tracking. Advanced analytics will support administrators in predicting course demand, tracking enrollment patterns, and identifying at-risk students early. This system will enforce role-based access control and log all data access to ensure compliance with FERPA regulations.

Target Audience

The platform serves multiple stakeholders across GSU:

- Students seeking an accessible and mobile-friendly course registration system without experiencing any errors.
- Advisors who need tools to plan student schedules, manage special circumstances, and monitor student progress.
- Faculty and Department Administrators requiring accurate enrollment, waitlist, and course capacity data.
- University Leadership and IT Services who require 99.99% uptime for university operations and compliance tracking.

Benefits

The new registration system will deliver:

- Enhanced Student Experience: Reduced average course registration time per student from previous years baseline, provide planning and advising tools for upcoming course registrations, and receive real-time notifications under 5 secs.
- Operational Efficiency: Reduce manual override processing time by 50% from previous years.
- Accessibility and Inclusion: Support for students with disabilities and international students with WCAG 2.1 AA compliance and support for at least 5 high-demand languages.
- Data-Driven Insights: Analytics to inform academic planning and identify at-risk students.
- Scalable, Secure Infrastructure: Reliable performance during peak periods with secure, FERPA-compliant data practices.

Strategic Allignment

This solution aligns with GSU's strategic goals by improving student success, promoting inclusivity and accessibility, enhancing operational efficiency, and supporting data-driven decision-making. It allows the university to meet the demands of a growing, diverse student body by modernizing their technology to support long-term growth and academic success.

Objectives and Key Results

Objective 1: Successfully deploy the Course Registration System

1. Complete system deployment on cloud infrastructure by the start of the next registration term.
2. Integrate all core systems with all existing university systems (SIS, LMS, Financial Aid, DAS, etc.) before go-live.
3. Resolve 95% of Priority 1 defects (system outages, login failures) deployment bugs within 48 hours of go-live.

Objective 2: Ensure real-time data accuracy and completeness

1. Verify that course availability, prerequisites, and enrollment data are updated within 2 seconds during normal load and under 5 seconds during peak load
2. Ensure waitlist notifications within 2 seconds with 99% reliability.
3. Reduce manual data correction requests from advisors and administrators by 50%.

Objective 3: Strengthen Administrative Control and Insights

1. Provide dashboards to all department administrators with real-time dashboards for course capacities, waitlists, and enrollment patterns.

2. Implement predictive analytics to identify at least 95% of at-risk students during registration periods.
3. Manual course override and seat adjustment process by 50% from previous year.

Objective 4: Continuously improve the Course Registration System based on user feedback

1. Conduct user surveys before and after registration window with at least 500 student and staff participants.
2. Achieve at least 90% user satisfaction in post-update feedback surveys.
3. Reduce reported usability issues by 75% within the first year after deployment.

Objective 5: Accommodate the growing number of users

1. Maintain system performance while supporting 20% annual growth in student enrollments.
2. Add additional server resources to support a 30% increase in peak load.
3. Successfully handle 10,000+ concurrent course search requests without failures.

Objective 6: Ensure all students have fair access to the course registration system

1. Reduce use of third-party registration tools by 90%.
2. Achieve less than 1% registration errors due to system bottlenecks or concurrency issues.
3. Ensure at least 95% of students, including international and disability-access students, can complete registration independently.
4. Provide automated conflict resolution for scheduling overlaps for at least 95% of students.

Objective 7: Ensure system performance during peak registration periods

1. Achieve 30% increase in successful registrations during peak periods compared to the previous system.
2. Maintain system uptime of at least 99.99% during peak registration periods.
3. Ensure all course search queries return results less than 5 seconds under peak load conditions.

Roles and User Needs

Role 1: Students (UG, Graduate, Full-Time, On-Campus, Online, and Degree-Seeking)

Purpose & Goals

- Register for required and elective courses within designated enrollment windows without system errors.

- Track degree progress and prerequisites clearly.
- Maintain enrollment status for financial aid, visas, and graduation eligibility.

Skills & Requirements

- Ability to log into web systems, search databases, and complete online forms independently.
- Ability to navigate web-based systems.
- Varying accessibility needs (screen readers, multilingual support).

Work Activities & Tasks

- Search and enroll in courses.
- Review prerequisites and degree requirements.
- Monitor enrollment status and waitlists.
- Adjust schedules during add/drop periods.

Task Characteristics

- Frequency: High during registration periods.
- Collaboration: Mostly individual.
- Time Criticality: Very high (registration windows, deadlines).
- Risk: High. Errors can delay graduation or impact financial aid/visa status.
- Information Used: Course catalog, degree requirements, enrollment status.
- Multitasking: Common (students often compare multiple courses simultaneously).

Needs from the System

- Real-time course availability and prerequisite validation.
- Clear degree progress visualization.
- Transparent registration queue order and timestamp-based seat allocation.
- Interface compliant with WCAG 2.1 AA accessibility standards.
- Notifications for deadlines and enrollment changes.

Role 2: Academic Advisors

Purpose & Goals

- Guide students academically and professionally.
- Identify enrollment risks early.
- Support course planning and graduation readiness.

Skills & Requirements

- Strong interpersonal communication.
- Knowledge of degree programs and policies.

- Organizational and analytical skills.

Work Activities & Tasks

- Conduct advising sessions.
- Review student enrollment and academic progress.
- Assist with overrides and special permissions.

Task Characteristics

- Frequency: Regular.
- Collaboration: With students, faculty, and compliance offices.
- Time Criticality: High during registration and advising cycles.
- Risk: Moderate. Incorrect guidance affects student outcomes.
- Information Used: Student records, enrollment data, degree audits.
- Dedicated role during work hours.

Needs from the System

- Centralized, real-time student data.
- Enrollment risk indicators.
- Easy access to degree progress and class capacity.
- Tools to initiate overrides and advising workflows.

Role 3: Faculty

Purpose & Goals

- Support students in course planning and academic progress.
- Monitor enrollment levels and manage course capacity.
- Approve overrides and provide timely academic guidance.
- Identify students at risk of delayed graduation.

Skills & Requirements

- Knowledge of degree requirements and course prerequisites.
- Strong communication and advising skills.
- Familiarity with institutional academic policies.
- Ability to use digital advising and registration tools.

Work Activities & Tasks

- Conduct advising sessions with students.
- Review student enrollment status and academic progress.
- Approve course overrides and special permissions.
- Monitor class enrollment and waitlists.

- Communicate changes or requirements to students.

Task Characteristics

- Frequency: Regular, with peaks during registration periods.
- Collaboration: With students, IT staff, and compliance offices.
- Time Criticality: High during registration and advising cycles.
  Risk: Moderate. Incorrect guidance or delayed approvals can affect student outcomes.
- Information Used: Student records, enrollment data, degree audits, course rosters.
- Dedicated to tasks during working hours.

Needs from the System

- Real-time visibility into student enrollment and degree progress.
- Easy tools for approving overrides and managing waitlists.
- Centralized access to course capacity and prerequisite information.
  Alerts for at-risk students or enrollment issues.

Role 4: IT Department

Purpose & Goals

- Maintain 95% uptime and support 5,000+ concurrent users during peak registration.
- Ensure data security and regulatory compliance.
- Support continuous improvement of the registration platform.
- Minimize system downtime and errors during peak registration periods.

Skills & Requirements

- Technical expertise in scalable systems design and architecture.

- Knowledge of cybersecurity and data privacy regulations.
- Troubleshooting and system monitoring skills.
- Ability to communicate technical issues to non-technical stakeholders.

Work Activities & Tasks

- Monitor system performance and availability.
- Manage authentication and access controls.
- Implement updates, patches, and system enhancements.
- Respond to incidents and resolve outages.
- Collaborate with academic and administrative teams on system requirements.

Task Characteristics

- Frequency: Ongoing.
- Collaboration: With advisors, registrar staff, and security teams.

- Time Criticality: Very high during system outages or peak usage.
- Risk: High. Failures or breaches impact the entire university.
- Information Used: System logs, performance metrics, security alerts.
- Dedicated role during working hours.

Needs from the System

- Monitoring dashboards and automated alerts.
- Scalable infrastructure to handle peak registration loads.
- Strong authentication and encryption mechanisms.
- Administrative tools for system configuration and maintenance.
- Detailed audit logs for compliance and troubleshooting.

Benefits and Features

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1. Integrated Screen Readers

Description: Accessibility features integrated into the registration system that is fully compatible with screen readers and other assistive technologies, so visually impaired students can navigate and complete registration using audio guidance and keyboard controls.

Benefit: Allows students who are visually impaired to complete registration independently without staff assistance. It reduces barriers caused by unreadable buttons or missing/hidden labels. This inclusion of accessibility features also reduces the need for manual staff assistance and supports compliance with accessibility standards and disability regulations.

2. Mobile-Friendly Registration Interface

Description: The registration system is designed to work smoothly on phones and tablets with intuitive interfaces and buttons that are easy to tap. Students can search for courses, build schedules, and register from a mobile device without needing a desktop computer or laptop.

Benefit: Allows students to register more easily and quickly from anywhere without switching between multiple windows or needing a laptop/PC, which reduces session abandonment rates during peak registration periods and supports students who primarily rely on mobile devices.

3. Multi-Factor Authentication Services

Description: Using the system requires approval on another device to authenticate the correct user and confirming their identity using a second method of a phone code or authentication app. This ensures that only authorized users can access registration and student records.

Benefit: Prevents unauthorized access to student academic and financial records. It increases trust in the system for students, faculty, and administrators while supporting institutional security and compliance requirements. It also lowers security risk exposure.

4. Integrative Degree Progress Tracker

Description: The system displays a real-time visual dashboard showing completed, in-progress, and remaining degree requirements.

Benefit: Gives students clear visibility into graduation progress and eliminates the need for students to manually track requirements across multiple systems. Additionally, this reduces advisor time spent manually verifying degree completion requirements as students can check their progress independently.

5. Automatic Scaling Capabilities

Description: The system has the capability to automatically increase capacity during high-demand periods and scales down after, maintaining fast response times even under heavy use.

Benefit: Automatic scaling ensures the registration system stays responsive during high-demand periods by automatically adding system capacity when demand spikes and reducing it when traffic drops. Students would experience fewer slowdowns and outages and the IT department avoids emergency interventions.

6. Real-Time Waitlist Updates

Description: Course seat availability and waitlist positions update automatically as changes occur so students can always see current course information when making decisions. This includes push notifications when seats open.

Benefit: Reduces refresh requests and duplicate system queries for seat availability and course waitlist requests, which reduces system overloads and helps students register more effectively.

7. Conflict and Risk Alerts

Description: The system automatically checks a student's schedule during registration and shows clear warnings for issues like time conflicts, missing prerequisites, or compliance risks before submission. Alerts are written immediately so problems can be fixed early.

Benefit: Students avoid preventable registration errors, while advisors and compliance offices spend less time on corrections or manual advising. This improves registration accuracy and supports smoother operations during peak periods as well as reduces confusion and frustration for students.

8. Course Analytics Dashboard

Description: The system includes built-in reporting dashboards that displays enrollment trends by department, seat fill rates, and historical demand comparisons. Instead of pulling data from multiple systems, authorized users can see visual summaries and reports in one place.

Benefit: Academic departments can make faster decisions about course offerings and resource allocation with a centralized system, improving overall organizational efficiency and transparency. This reduces manual reporting and gains more accurate insights.

Non-Functional Requirements and System Constraints

Security Needs

NFR_001: Security and Data Protection

Data Protection

- Encryption in transit and at rest: All sensitive data (PII, financial, immigration, disability records) will be encrypted when transmitted and when stored in databases and backups to prevent interception or theft if any of the systems are compromised.
- Data minimization and masking: Sensitive data and fields (SSN, financial aid or immigration details, disability accommodations) will be masked in UI views for non-admin roles and redacted in application logs by default in the user interface and logs will only be visible to authorized users.
- Audit logging: Security events (logins, permission changes, overrides, etc.) will be logged to support incident responses and investigations.

User Authentification

- Single Sign-On (SSO): Users will authenticate using the university's identity provider to reduce password reuse and to improve account control.
- Multi-Factor Authentication (MFA): MFA will be required to authenticate users during high-risk actions such as personal data updates, transcript access, financial record access.

- Session management: Sessions will automatically expire after inactivity and require re-authorization for sensitive actions. This prevents simultaneous logins to reduce account takeover risks.
- Audit logging: All actions are recorded with user ID, timestamps, and action type to ensure accountability and to support incident responses.

Security Roles

Permissions	Students	Faculty	Academic Advisors	Accessibility Services	Govt. Compliance	IT Dept.	Security
View class availability	✓	✓	✓	✓	✓	✗	✗
Search courses	✓	✓	✓	✓	✓	✗	✗
View course details	✓	✓	✓	✓	✓	✗	✗
View own schedule	✓	✓	✓	✓	✓	✗	✗
Register/drop classes	✓	✗	✗	✗	✗	✗	✗
Join waitlists	✓	✗	✗	✗	✗	✗	✗
View class rosters	✗	✓	✓	✗	✗	✗	✗
View degree progress	✓	✓	✓	✗	✗	✗	✗
View accommodation flags	✓	✗	✓	✓	✗	✗	✗
View aid/visa compliances	✓	✗	✓	✗	✓	✗	✗
Submit override requests	✓	✓	✓	✗	✗	✗	✗
Approve overrides	✗	✓	✓	✗	✗	✗	✗
Run enrollment + compliance reports	✗	✗	✓	✓	✓	✗	✓
Access system performance data	✗	✗	✗	✗	✗	✓	✓
Manage integrations	✗	✗	✗	✗	✗	✓	✓

Manage roles and permissions	✗	✗	✗	✗	✗	✗	✓
View audit logs	✗	✗	✗	✗	✗	✗	✓

Scalability Needs

NFR_002: Scalability and Growth Accomodation

Description: The course registration system must scale to support GSU's large multi-campus population and a growing student and staff user base without compromising its performance or architectural designs.

Rationale: GSU serves 45,000 students across multiple campuses with 4,800 course sections per term and 300+ academic programs. Registration demand is highly concentrated into short peak windows, creating extreme concurrency loads. With continued enrollment growth and program expansions across 15 colleges and schools, the system must scale their user base and data volume without performance degradation.

Metrics

- Support 45,000+ active students and 3,200+ faculty/staff users
- Support 20% annual growth in user base
- Handle 30% increase in peak loads
- Linear cost scaling with user and data growth

Acceptance Criteria

- System capacity can be increased without service interruptions or downtime
- Load testing at 130% of current peak shows no critical failures
- New programs, courses, and departments can be added without architectural redesigns
- Maintain performance with 20% annual growth in enrollment and audit data

Availability Needs

NFR_003: Availability Continuity

Description: The course registration system must maintain very high availability, especially during peak registration periods, to ensure users can complete and access registration actions within a time-sensitive window without disruptions.

Rationale: The system is used 24/7 by all users, meaning it's important the system is running at all times with the least amount of downtime possible to prevent inconveniences between users. Even short outages can prevent or delay compliances and create institutional risks.

Metrics

- Recovery Time Objective (RTO): < 2 hours
- Recovery Point Objective (RPO): < 5 minutes
- System has a 99.99% uptime during registration periods
- System has a 99.9% uptime during normal hours

Acceptance Criteria

- All scheduled maintenances are during off-peak hours
- Monitoring reports confirm uptime meets 99.9% peak and 99.9% normal targets
- System remains operational during single-node failures
- Recovery time stays within RTO targets

Interoperability Needs

NFR_004: System Integration

Description: The registration system must seamlessly integrate with other GSU enterprise platforms.

Rationale: Accurate registration operations require real-time data exchange on all systems. Without reliable integration, the university could face compliance errors, delayed updates, duplicate data entry, or inconsistent student records.

Required Integrations

- SIS: student records, enrollment status, academic standing
- LMS (Canvas): course rosters and section enrollments
- Financial Aid System: enrollment load and aid eligibility
- Identity Management: authentication and roles
- Payment System: registration-related payment status
- Degree Audit: degree progress data
- Academic Planning: planned courses and prerequisites
- Room Management: course sections and room assignments

Metrics

- Integration uptime ≥ 99.9%
- Cross-system data accuracy ≥ 99.5%
- API error rate ≤ 1%
- Automated sync success rate ≥ 99%

Acceptance Criteria

- Enrollment changes automatically propagate to SIS, DAS, and APS
- Course registrations automatically update Canvas rosters
- Financial aid eligibility updates after enrollment changes
- Compliance indicators update without manual re-entry
- No duplicate manual data entry across integrated systems

Sustainability Goals and Needs

NFR_005: Sustainability and Resource Efficiency

Description: The registration system must be designed and deployed to minimize consumption and resource usage through efficient and optimization practices.

Rationale: GSU has institutional sustainability goals and operates at a large scale (45,000 students, high peak loads). Efficient infrastructure design reduces environmental impact and operational cost while maintaining performance and availability.

Metrics

- Use cloud or data center providers with documented green / renewable energy commitments
- Enable auto-scaling so compute resources scale down during off-peak periods
- Average compute utilization ≥ 60% (avoid over-idling)
- Resource expansion for peak-only periods only

Acceptance Criteria

- System supports automatic scale up/down based on load
- Off-peak resource usage demonstrably lower than peak usage
- Deployment hosted on energy-efficient or sustainability-certified infrastructure
- Sustainability metrics included in infrastructure monitoring reports

Standards and Compliance Needs

NFR_006: Regulatory and Accessibility Compliance

Description: It is imperative that the course registration system complies with all standards and regulations as it handles sensitive student academic and personal data, which must have equitable access for all users.

Data Security

- FERPA compliance
    - Protect student confidentiality of education records
    - Restrict access based on authorized roles

    - Maintain access audit trails
- GDPR compliance for international students
    - Protect personal data for users
    - Support data minimization and purpose limitation

Accessibility

- Full compliance with WCAG 2.1 AA standards:
    - Screen readers
    - Keyboard navigation
    - High contrast modes
    - Text scaling
    - Alternative text for all images
    - Multiple language support

Prioritization and Road Map

Release 1 - MVP (Core Registration Platform)

Timeframe: Months 0 - 3

Goals

- Establish foundational system infrastructure.
- Enable basic course registration workflows.
- Ensure accessibility and security compliance.
- Reduce registration errors and system downtime.

Key Features & Capabilities

- Unified course search and enrollment
- Real-time seat availability
- Prerequisite validation
- ADA-compliant interface
- Role-based authentication (students, advisors, IT)
- Basic advisor dashboard
- Secure student profiles

Success Metrics

- ≥90% successful registrations
- <1% system error rate
- 99.5% uptime during registration

- Reduce manual registration override corrections by 25% compared to 2024 baseline

Supporting Tasks

- Data migration from legacy systems
- Accessibility audit and remediation
- Security configuration and role setup
- Stakeholder onboarding
- User acceptance testing

Release 2 - Operational Enhancements

Timeframe : Months 4 - 6

Goals

- Improve advising efficiency.
- Enhance system performance and usability.
- Reduce manual administrative workload.

Key Features & Capabilities

- Degree progress visualization
- Override workflows
- Automated registration notifications
- Enrollment risk indicators
- Reporting dashboards
- Performance optimization

Success Metrics

- 30% reduction in advisor manual workload
- 25% faster override approvals
- Post-registration survey (n500) shows 95% satisfaction rating of 4 or higher (5-point scale)."

Supporting Tasks

- Advisor training sessions
- Workflow refinement
- Load and stress testing
- Documentation updates

Release 3 - Optimization & Analytics

Timeframe: Months 7 - 9

Goals

- Optimize user experience.
- Introduce predictive insights.
- Automate compliance monitoring.

Key Features & Capabilities

- Predictive enrollment alerts
- Advanced analytics dashboards
- Personalized course recommendations
- Compliance automation tools

Success Metrics

- Reduce course bottleneck-related graduation delays by 15% within two academic years.
- 50% decrease in misinformation errors
- 99.99% uptime during peak periods

Supporting Tasks

- Impact analysis
- Model tuning
- Stakeholder feedback incorporation
- Institutional reporting

Approach to Prioritization

Capabilities were prioritized using the following criteria:

- **Student impact first:** Features that reduce registration friction were prioritized earliest.
- **Dependency-based sequencing:** Core infrastructure precedes analytics and automation.
- **Risk reduction:** Security, accessibility, and system stability were addressed in early releases.
- **Stakeholder value:** Advisor and compliance workflows follow student-facing essentials.
- **Incremental value delivery:** Each release provides usable functionality rather than waiting for full system completion.

Rationale for Feature Priority

The MVP focuses on foundational infrastructure because secure authentication, accessibility compliance, and reliable enrollment workflows are prerequisites for all advanced capabilities. Operational enhancements address advisor efficiency and system performance, directly targeting

the largest operational pain points identified in stakeholder research. Optimization features such as predictive analytics are introduced last, once sufficient enrollment data exists to support meaningful insights. This phased approach enables early value delivery, supports iterative improvement, and reduces project risk while progressing toward a fully integrated, scalable registration platform.

Risks, Assumptions, and Dependencies

Risks

1. Performance During Peak Registration

Description: Registration periods create extremely high demand. If the system cannot handle this volume, it may slow down, freeze, or crash completely. This could delay enrollment, create stress, and damage trust in the new platform.

Mitigation: Perform extensive stress testing that simulates usage beyond expected peak levels. Use scalable infrastructure that can automatically increase capacity during high-demand periods. Monitor performance continuously during registration and have a rapid response plan in place if issues arise.

2. Modernization Risk: Incompatibility

Description: The new registration system must connect with several existing university systems, including student records, financial aid, payments, and learning platforms. If these systems do not communicate properly, information may not update correctly. Even small mismatches in data formats or timing could create confusion for students and staff.

Mitigation: Test integrations early and run both systems in parallel before full launch. Establish clear points of contact with each service provider to resolve issues quickly.

3. Modernization Risk: Data Security

Description: The registration system contains sensitive student data, unauthorized access or data breaches could lead to legal consequences, regulatory penalties, and reputational harm to the university.

Mitigation: Require multi-factor authentication for users, encrypt all sensitive data, and limit access based on user roles. Conduct independent security reviews and regular system audits. Provide ongoing training to staff about data protection responsibilities.

4. Modernization Risk: Data Migration

Description: Transferring historical data from the old system into the new platform presents a risk of incomplete or inaccurate records. Errors in course history, prerequisites, or enrollment status could affect graduation progress and financial aid eligibility.

Mitigation: Clean and validate data before migration begins. Migrate information in stages rather than all at once. Perform detailed record comparisons between systems and require formal approval from key departments before full launch.

5. Project Management Risk: Project Delays

Description: There is a risk that additional feature requests or coordination challenges could extend timelines. Delays may conflict with academic calendar deadlines and increase costs.

Mitigation: Clearly define project scope at the outset and prioritize features by phase. Establish regular milestone reviews and decision checkpoints. Limit changes during critical development stages unless they are essential for compliance or system stability.

Assumptions

Business Assumptions

- The university leadership supports modernization of the registration system and will provide sufficient funding.
- The projected implementation and maintenance costs are within the approved budget.
- Stakeholders (students, faculty, advisors, administrators) will adopt and use the new system.
- The new system will provide measurable improvements over the current legacy platform.
- Service providers will corporate during integration work

Project Management Assumptions

- The project can be completed within the defined academic timeline.
- Stakeholders will be available for interviews, feedback sessions, and user testing.
- Internal teams and external providers will meet agreed-upon milestones.

Technological Assumptions

- Existing university systems support integration with the new platform.
- The system can scale to meet peak registration demand.
- Data migration from the old system is technically feasible.
- Required security measures can be implemented successfully.
- Most registration errors are caused by unclear interface design rather than user incompetence, meaning improved UI/UX will reduce error rates.

Compliance and Security Assumptions

- The system will comply with FERPA and other applicable privacy regulations.
- Accessibility requirements (WCAG 2.1 Level AA) can be fully met.
- Multi-factor authentication and role-based access controls will be supported by existing identity systems.
- Security audits and compliance reviews will be completed before launch.

User Assumptions

- Students' perception of fairness in registration depends on consistent system behavior across devices and contexts.
- Students prefer transparent waitlist positions and clear prerequisite verification.
- Users have access to reliable internet and a computer or mobile device.
- Faculty and advisors are willing to transition from manual processes to automated workflows.
- Clear in-system guidance will reduce confusion and decrease the need for manual interventions.

Dependencies

- Student Information System (SIS) for accurate student records, transcripts, and prerequisite validation.
- Financial Aid System for real-time enrollment and eligibility updates.
- Identity Management System for secure authentication and role-based access.
- Payment Processing System for tuition assessment and financial hold resolution.
- Learning Management System (LMS) for automatic course access after registration.
- Degree Audit System for tracking academic progress and requirements.
- Scalable Hosting Infrastructure to support peak registration demand and uptime targets.
- Reliable System Integrations (APIs) to ensure real-time data exchange across platforms.
- University IT and Security Teams for monitoring, maintenance, and compliance oversight.
- Stakeholder participation for testing, feedback, and adoption.

Scope of Solution

In Scope

This project includes the design and implementation of a modern, real-time university course registration. The overall deliverable of this project is to design and implement a modern, web-based, and mobile-responsive Course Registration System for Global State University. The system will serve as a centralized platform for managing course enrollment, academic planning, and registration-related processes while integrating with existing university systems. In line with this objective, activities and deliverables that fall within scope include those that allow users to:

- Search for courses using filters such as department, program, time, delivery format, and instructor; view real-time seat availability; and access prerequisite and degree requirement information.
- Register for courses, manage waitlists with transparent position tracking, receive automated notifications about capacity changes and registration windows, and build multiple schedule scenarios using a planning "shopping cart."
- Automatically verify prerequisites and enrollment eligibility prior to registration to reduce errors and compliance risks.

In addition, to ensure the system meets institutional needs and performance expectations, the following technical and operational elements are included in scope:

- Development of a mobile-responsive, accessible interface compliant with accessibility standards
- Integration with existing university systems, including student records, financial aid, payment processing, identity management, degree audit, and learning management systems
- Performance optimization to support peak registration periods with high concurrent usage and defined uptime targets
- Implementation of security controls, including role-based access, authentication safeguards, and data protection measures.

Out of Scope

Deliverables that fall outside of the scope will include

- Replacement or redesign of existing enterprise systems such as the Financial Aid System, Degree Audit System, etc.
- Guaranteeing course availability or resolving institutional capacity constraints
- Changes to institutional academic policies, grading standards, degree requirements, curriculum design, or faculty governance processes.
- Direct management of immigration processing, visa issuance, or external government compliance systems beyond tracking enrollment status.

Change Log

Current State

Summary of Feedback	Changes
Problem statement needed to be more specific about the course registration process	Refined language to explicitly focus on course registration workflows, peak enrollment periods, and student navigation challenges rather than general system issues.
Explain what is meant by “disconnected systems”	Clarified that enrollment, advising, accessibility, financial aid, and compliance systems operate independently, creating data silos and limiting cross-department communication.
More specification on compliance department	Added specific examples of compliance stakeholders, including the Financial Aid Office, International Student Office, Disability Services, and Registrar’s Office, and clarified how delayed or inaccurate registration data increases the risk of policy violations related to FERPA, visa requirements, and financial aid eligibility.
Clarify meaning of inconsistencies and their impact on transparency and efficiency	Added clarification that disconnected systems create data inconsistencies which reduce transparency and efficiency, requiring staff to manually reconcile information across platforms.

Stakeholders

Summary of Feedback	Changes
Needed clarification and explanation of why the system uptimes would be 99.99% throughout the year.	Changed from 99.99% system uptime throughout the year to 99.99% system uptime only during peak registration periods.
Needed clarification on the specifics of what government regulations needed to be followed, such as FERPA or distinguishing visa requirements etc.	Changed to be more specific on what exactly needed to be addressed in terms of FERPA, visa, and financial aid eligibility.
Needed clarification on the meaning on other	Changed to show what exactly were the

languages and also needed to be more specific and not mention something already mentioned before	accessibility requirements needed as well as needing to sustain multiple languages.

Research and Analysis Summary

Summary of Feedback	**Changes**
Needed example of how users struggle with the current system.	Added an example explaining how students keep multiple tabs open to manage the course registration process.
Citation of Mowreader's paper was too long.	Restructured the sentences introducing the paper to be more succinct.
Needed to establish a clear connection between the literature and the project.	Added a sentence explaining how Bano's paper aligns with our overall topic and how we will use it to develop our solution.
(In References) Needed to include all authors' names in the Bano citation in the reference section.	Added all of the authors' names to the citation.